

Diamond Price Prediction with Uncertainty Estimates

Final data science project: predict diamond prices (USD) from grading attributes and dimensions, with prediction intervals. Dataset: 53,940 diamonds (seaborn-data), 53,772 after cleaning.

Pipeline: data quality checks → EDA → feature engineering (volume, log-carat, ordinal grade encoding) → 3-model comparison (5-fold CV) → grid-search tuning → held-out test evaluation → error analysis by carat and clarity → quantile prediction intervals → permutation importance.

Results (test set, tuned HistGradientBoosting on log-price):

- R^2 (log scale) 0.992, MAE about \$268, RMSE about \$526, MAPE about 6.5%
- Ridge baseline CV MAE about \$443, so the tuned model cuts error by roughly 40%
- Nominal 90% prediction interval covers 85.1% of test diamonds (slightly under-covers)

Findings: size features dominate, then clarity and colour; cut adds little once size is known. Relative

error is highest for large stones and the I1 clarity grade.

Run

```
pip install numpy pandas matplotlib seaborn  
scikit-learn  
jupyter notebook  
diamond_price_prediction.ipynb
```

The notebook downloads the data from GitHub, so it runs in Google Colab without uploads. Full run takes several minutes.

Limitations

Historical snapshot, no certification-lab features, intervals not perfectly calibrated, not for real appraisal.